SAGNIK DATTA BHOWMICK

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EDUCATION

Institute of Engineering and Management, Kolkata 2025 – 2029
B.Tech in Computer Science and Engineering (2nd Year) | CGPA: 9.315/10

Indian Institute of Technology Madras Ongoing
BS in Data Science and Applications (Foundation Level, Term II) | Foundation I SGPA: 10/10

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript
ML / Computer Vision: PyTorch, Hugging Face Transformers, Transfer Learning & Fine-Tuning, Semantic Segmentation, OpenCV, MediaPipe, Scikit-learn, NumPy
Web & Backend: React, Vite, FastAPI, REST APIs, HTML, CSS, Axios
Data & Tools: Pandas, SymPy, Matplotlib, Git, GitHub, Linux, VS Code
Mathematics: Probability, Statistics, Linear Algebra, Calculus, Discrete Mathematics, Combinatorics

PROJECTS

Transfer Learning for Road Segmentation: DeepLabV3, FCN & SegFormer github.com/Sagn-2007
PyTorch, Hugging Face, KITTI
- Fine-tuned three pretrained architectures (SegFormer-B0/B2, DeepLabV3-ResNet50, FCN-ResNet50) for binary road segmentation on the KITTI Road dataset (231 train / 58 validation images, fixed seed), replacing classifier heads and comparing frozen vs. partially fine-tuned backbones.
- Fine-tuning ResNet layer4 lifted validation IoU from 25.0% to 84.9% (DeepLabV3) and 59.5% to 87.1% (FCN); progressive unfreezing of SegFormer-B0 stages 2–3 with layer-wise learning rates reached 90.27% IoU / 94.89% Dice.
- Scaled to SegFormer-B2 for 92.95% IoU / 96.34% Dice (+2.68 pts over B0); used best-validation-IoU checkpointing and per-image IoU visualizations; trained on a 6 GB RTX 3050 laptop GPU.

Road Segmentation using Fully Convolutional Networks github.com/Sagn-2007
PyTorch, OpenCV, NumPy
- Built a custom encoder-decoder FCN from scratch in PyTorch on the KITTI Road Dataset, with a custom Dataset class, end-to-end training/inference pipeline, checkpointing, batch prediction and mask overlay visualization.

Real-Time Spinal Posture Detection github.com/Sagn-2007
Python, OpenCV, MediaPipe, Scikit-learn
- Built a two-camera (front + side) posture monitor using MediaPipe pose landmarks; computed spine-angle features and trained a Random Forest classifier to flag deviations in real time with threaded capture. Top 20 at Innovacion Tech Fest.

Mathematical Graph Plotter: Full-Stack Web App github.com/Sagn-2007
React, Vite, FastAPI, SymPy, Pandas, Plotly
- Built a React/Vite + FastAPI app with REST endpoints returning Plotly-ready data; wrote a safe SymPy expression parser (no eval()), a Pandas CSV-upload pipeline, and a debounced (600 ms) UI supporting 10 overlaid functions with zoom, pan and PNG export.

Arduino-Based Fire Alarm System IEM IIFR Lab
Arduino Nano, Sensors, PCB
- Designed and assembled a fire-detection circuit with temperature and smoke sensors, a custom PCB and alert circuitry.

ACHIEVEMENTS

- WBJEE: Rank 4,110 of 101,643 (top ~4%) | CUET-UG 2025: 98.26 percentile (English), 97.07 (Physics), 96.48 (CS/IT), 94.76 (Mathematics)
- IISER Aptitude Test: qualified for admission (offer not availed) | AISSCE (Class XII): 91.8%
- Innovacion Tech Fest: poster on Anthropic's Constitutional AI research; posture-detection project in Top 20
- IEM Heals: poster on AI-driven de novo drug discovery, qualified for finals | NPTEL Joy of Computing using Python: top 5%

CERTIFICATIONS

- Python Data Structures, University of Michigan (Coursera), 97% | Advanced Relational Database and SQL (Coursera), 100%
- Build a CI/CD Pipeline with Docker (Coursera), 100% | Unsafe Code in .NET: Memory Management and Optimization (Coursera), 100%
- Introduction to Retrieval Augmented Generation (RAG), Duke University (Coursera), 80%